

Heatwave impacts on intertidal seagrass reflectance: from laboratory experiment to satellite mapping of seagrass heat shock index

Simon Oiry* Bede Ffinian Rowe Davies¹ Philippe Rosa¹
Augustin Debly¹ Maria Laura Zoffoli[†] Anne-Laure Barillé[‡]
Nicolas Harin³ Marta Román¹ Pierre Gernez¹
Laurent Barillé¹

2025-10-01

Seagrasses play a vital role in coastal ecosystems, providing habitat, stabilizing sediments, and contributing to carbon sequestration. However, climate change has increased the frequency and intensity of heatwaves, posing a significant threat to seagrass health. This study investigates the effects of marine and atmospheric heatwaves on the spectral reflectance of the intertidal seagrass *Zostera noltei*. Laboratory experiments were conducted under controlled heatwave conditions, where hyperspectral reflectance measurements were taken to assess the impacts over time. Heatwaves caused a substantial decline in seagrass reflectance, particularly in the green and near-infrared regions, corresponding to the browning of green leaves. Key vegetation indices, including the Normalized Difference Vegetation Index (NDVI) and Green Leaf Index (GLI), showed pronounced reductions under heatwave stress, with NDVI values decreasing by up to 34% and GLI by 57%. A novel metric, the Seagrass Heat Shock Index (SHSI), was developed to quantify the transition of seagrass leaves from green to brown, demonstrating a strong ability to capture the effects of heatwave exposure on seagrass coloration. Multispectral satellite observations corroborated the laboratory results, revealing widespread browning of seagrass leaves during marine and atmospheric heatwave events in South Brittany, France. Notably, darkened seagrass patches were observed in intertidal areas exposed to temperatures exceeding 32°C for over 13.5 hours per day. These findings

*Institut des Substances et Organismes de la Mer, ISOMer, Nantes Université, UR 2160, F-44000 Nantes, France

[†]Consiglio Nazionale delle Ricerche, Istituto di Scienze Marine (CNR-ISMAR), 00133 Rome, Italy

[‡]Bio-littoral, Immeuble Le Nevada, 2 Rue du Château de l'Eraudière, 44300 Nantes, France

highlight the potential of spectral reflectance as a tool for detecting early signs of heatwave-induced stress in seagrasses, offering a valuable method for remote sensing-based habitat assessment. The present study underscores the potential of remote sensing to capture rapid environmental changes in intertidal zones, enabling for continuous monitoring of seagrass meadows under the current and future climate regimes.

Introduction

Seagrasses play a crucial role in coastal ecosystems by providing habitats and feeding grounds for various marine species, supporting marine biodiversity, and contributing to primary production and carbon sequestration (Sousa et al., 2019; Unsworth et al., 2022). Seagrasses are essential for several ecological functions, such as sediment stabilization (Infantes et al., 2022) or eutrophication mitigation by consuming nutrients (Gladstone-Gallagher et al., 2018). This justifies their use as indicators of environmental changes due to their sensitivity to water quality variations (Zoffoli et al., 2021). The interactions between seagrass meadows and their associated herbivores further enhance the delivery of ecosystem services, including coastal protection, fisheries support and provision of habitat and resources for birds (Gardner and Finlayson, 2018; Jankowska et al., 2019; Unsworth and Butterworth, 2021; Zoffoli et al., 2023). Understanding and preserving seagrass is vital for maintaining the biodiversity and productivity of coastal regions (Ramesh and Mohanraju, 2020; Scott et al., 2018).

Despite their crucial role in marine ecosystems, seagrasses face numerous threats that compromise their health and functionality. Intertidal seagrasses are subjected to a combination of aquatic and aerial conditions linked with tidal cycles, and face disturbance from terrestrial and aquatic stressors. Coastal development and human activities are primary threats, reducing the available habitat for seagrasses and increasing water turbidity, limiting light penetration, and photosynthesis (Waycott et al., 2009). Seagrasses are also threatened by runoff from agricultural fields and urban areas leading to nutrient enrichment. Eutrophication promotes the growth of seaweed in coastal waters, causing macroalgal blooms that compete with seagrasses for light and nutrients (Brun et al., 2003; Oiry et al., 2024; Thomsen et al., 2023). Pollution from industrial and agricultural sources introduces harmful chemicals and heavy metals into coastal waters, posing toxic risks to seagrass health (Bastos et al., 2023; Green et al., 2021; Zahoor and Mushtaq, 2023). Among manifold anthropogenic stressors, heatwaves (HWs), exacerbated by climate change, pose a severe threat to seagrasses, with catastrophic dieback events observed worldwide (Carlson et al., 2018; Marbà and Duarte, 2010; Moore and Jarvis, 2008; Strydom et al., 2020; Thomson et al., 2015).

Marine Heatwaves (MHWs) are defined by Hobday et al. (2016) as prolonged discrete anomalously warm water events, while Atmospheric Heatwaves (AHW) are defined by Perkins and Alexander (2013) as periods of at least three consecutive days with temperatures exceeding the 90th percentile of a time series covering at least 30 years. Subtidal seagrass meadows are exposed to MHWs, whereas at the interface between land and ocean, intertidal seagrasses are exposed to both MHWs and AHWs. HWs profoundly impact seagrass physiology, with

effects varying between species and geographic location. Widespread seagrass species such as *Zostera marina* exhibits high susceptibility to elevated sea surface temperatures during winter and spring, leading to advanced flowering, high mortality rates, and reduced biomass (Sawall et al., 2021). Similarly, *Cymodocea nodosa* shows increased photosynthetic activity during HWs but suffers negative effects on photosynthetic performance and leaf biomass during recovery (Deguette et al., 2022). Additionally, different populations of *Zostera marina* along the European thermal gradient exhibit varied photophysiological responses during the recovery phase of HWs, indicating differential adaptation capabilities among populations (Winters et al., 2011). High-latitude populations exhibited prolonged declines in photophysiological performance even after temperatures returned to control levels, whereas the low-latitude Adriatic population showed full recovery (Winters et al., 2011). These events intensify other stressors, such as overgrazing and seed burial, compromising recruitment (Guerrero-Meseguer et al., 2020). Although extensive research exists on marine heatwaves' effects on subtidal seagrasses (Arias-Ortiz et al., 2018; Deguette et al., 2022; Strydom et al., 2020), less attention has been given to intertidal habitats and even less to the effect of atmospheric extreme events on intertidal seagrass. Nonetheless, recent research showed that the low tide exposure of *Zostera noltei* to a simulated four-day atmospheric HW caused significant decreases in its photosynthetic efficiency, resulting in leaf necrosis and decay (Román et al., 2023).

The increased occurrence of extreme climate events calls for the implementation of monitoring strategies able to provide detailed and spatially explicit assessments of HWs effects on seagrass meadows. In such context, remote sensing, whose ability to map seagrass distribution over a variety of spatio-temporal scales has been demonstrated (Davies et al., 2024a, 2024b; Oiry et al., 2024; Román et al., 2021), proved useful to study the changes in seagrass coverage caused by extreme HW event (Strydom et al., 2020). The pigment composition of plants, such as chlorophylls, carotenoids, and anthocyanins, significantly influences their spectral signature in the visible range due to their specific light absorption properties (Davies et al., 2023; Douay et al., 2022; Olmedo-Masat et al., 2020; Ustin and Jacquemoud, 2020). During the senescence phase of seagrass' life-cycle, the degradation of chlorophyll and the unmasking of accessory pigments result in noticeable changes in leaf coloration and reflectance, including increased reflectance in the red and green wavelengths and shifts in the red-edge position (Boyer et al., 1988; Mariën et al., 2019; Peñuelas et al., 2004). Leaf browning, often observed after stress events, produces reflectance changes similar to those caused by senescence, enabling the detection of vegetation stress through remote sensing (Boyer et al., 1988; Peñuelas et al., 2004). Spectral indices such as the Brown Pigment Index (BPI) and the Photochemical Reflectance Index (PRI) have been developed to assess changes in terrestrial plant physiological status, including oxidative and drought stress (Garbulsky et al., 2011; Skendzic, 2023). While these effects are well-documented in terrestrial plants, the spectral reflectance changes associated with senescence and stress events such as MHWs or AHWs remain poorly studied on intertidal seagrasses.

This study aims to experimentally test the hypothesis that HWs alter the reflectance of the intertidal seagrass *Zostera noltei*. Controlled experiments in intertidal chambers were conducted to evaluate the direct impact of heat stress on seagrass reflectance. The findings will then be

applied to satellite remote sensing images, providing critical insights into the spatial extent and temporal dynamics of HW effects on seagrass meadows. By linking experimental results with large-scale observations of seagrass leaves' browning, the study aims to underscore the potential of remote sensing to enhance our understanding of seagrass responses to extreme thermal events across diverse settings and timescales.

Material & Methods

1 Laboratory Experiment

1.1 Sampling and acclimation of seagrasses

Seagrass samples were taken in summer 2024, at low tide, from a *Zostera noltei* (dwarf eelgrass) meadow located in Bourgneuf Bay, France (46°57'32.0"N, 2°10'37.0"W). A metal coring device was used to sample seagrass from an area of 30x15x5 cm (length x width x depth, respectively), maintaining the sediment structure and avoiding damage to seagrass rhizomes and leaves (Figure ?? A). These coring device enabled the collection of sediment samples at a consistent depth, minimizing variability between samples. A total of six samples were collected. Samples including seagrass, sediment, meiofauna, and macrofauna, were placed in plastic trays. Keeping the entire biota allowed for natural interactions between components and reduced stress on the seagrass. Seawater was added to each tray to avoid hydric stress caused by insufficient moisture during transportation (1h drive from the laboratory). Simultaneously, seawater was sampled 4km away from the seagrass sampling site and transported to the lab, where it was filtered using a 0.22 μm nitrocellulose filter to remove suspended particulate matter. The filtered seawater was used in the acclimation tank and the intertidal chambers. The seagrasses were acclimated for one week with a water temperature of 17°C, matching the *in situ* temperature during sampling, and a photosynthetically active radiation (PAR) of 150 $\mu\text{mol.s}^{-1}.\text{m}^{-2}$.



Figure 1: Illustration of the experiment. A: Seagrass field sampling using a coring device; B: Intertidal chamber used during the experiment; C: Seagrass sample inside a chamber during the experiment at high tide; D: Treatment sample at the start of the experiment; E: Treatment sample at the end of the experiment, 3 days after the start of the HW event.

1.2 Experimental design

A tidal cycle (i.e. regularly alternating 6h of low-tide and 6h of high-tide) was simulated in the laboratory using an intertidal chamber system from ElectricBlue® (Figure ?? B ; Electric Blue

(2023)). The experimental setup allowed only two tidal states: high tide or low tide, with no intermediate stages. The transition between these states took about 15 minutes to complete after initiation. During the phase of high tide, a volume of 30 L of filtered seawater was pumped and circulated through the chamber (Figure ?? B, C). During low tide, the seagrass sample was emerged. The acclimated seagrasses were split into two subsets and placed in two independent chambers used in parallel, with one chamber used for control and the other for the experimental treatment. The intertidal chambers were equipped with LED lights that emitted a low amount of red and infrared radiation. To achieve a Photosynthetically Active Radiation (PAR) intensity of up to $400 \text{ mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$, a filament bulb was added inside the chambers. During the diurnal phase of the experiment, the PAR was kept constant in both intertidal chambers. To follow the circadian cycle, these lights (both LED and filament bulb) were turned on and off each day, at the time of sunrise and sunset, respectively.

Air temperature and water temperature were controlled inside the experiment chambers in order to reproduce the range of variability observed in the field (Figure ??). Field temperature was measured using *in situ* sensors (T7.3 EnvLoggers from ElectricBlue®) deployed at the sampling site in August 2024. The loggers were positioned along a transect from the upper to the lower intertidal zone, attached to pre-existing wooden poles at the sediment surface. In complement, the daily temperature maxima recorded *in situ* were compared with measurements from the nearest Météo France weather station (Annexe A1, Section ??). The control chamber was kept at temperatures representing typical summer conditions, with water temperatures at 18°C and air temperatures from 19°C to 23°C, following natural daily temperature fluctuations (Figure ??). In the treatment chamber, the air temperature was adjusted to mimic an AHW that affected the seagrass meadow in Quiberon, South Brittany, France (47°35'40.0"N, 3°07'30.0"W), from September 2 to September 6, 2021. Air temperature in the experimental chamber was set to vary from 23°C (at night) to 35°C (daytime) during the first day of the experiment, and increase by 1°C daily during three consecutive days. Water temperature in the experimental chamber was also adjusted to mimic MHW conditions, starting at the seasonal baseline (18°C) and rising incrementally by 0.5°C daily to simulate the increasing temperatures during the event. This aimed to reproduce the thermal stress experienced by the seagrass meadow during a MHW (Figure ??). The experiment, with both treatment and control chambers, was repeated three times to obtain replicates (hereafter referred to as "Run").

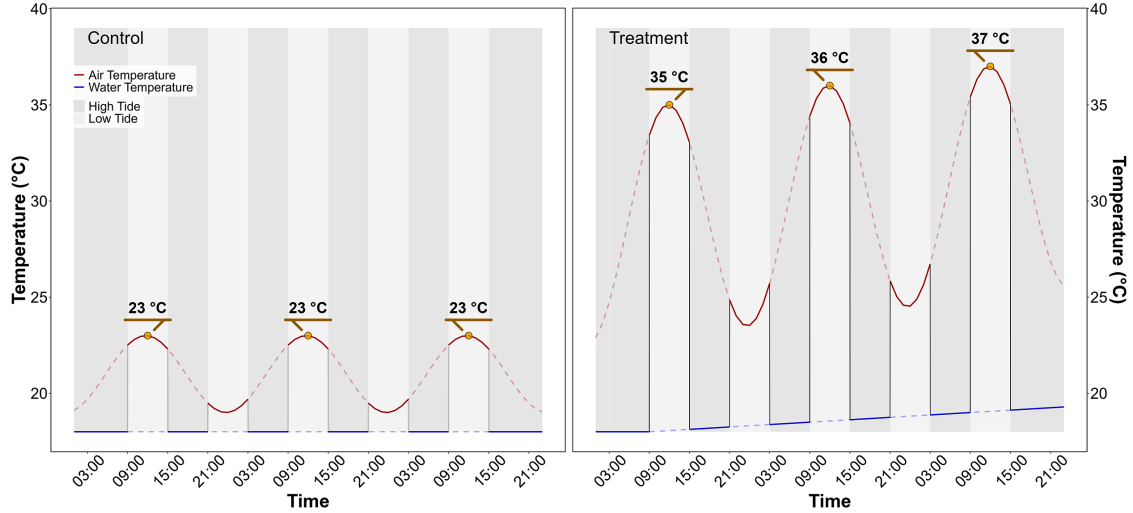


Figure 2: Temperature variation in the control (left) and treatment (right) intertidal chambers, during the HW experiment. The red line indicates air temperature, and the blue line water temperature. Due to the tidal cycle of immersion / emersion, the seagrasses experienced the temperatures represented by solid lines.

1.3 Optical measurements

1.3.1 Hyperspectral reflectance measurements

Throughout the experiment, the hyperspectral reflectance, $R(\lambda)$, of both the control and treatment seagrasses was measured using an ASD HandHeld 2 equipped with a fiber optic extension placed inside the chamber. The measurement set up made it possible to automatically acquire $R(\lambda)$ without opening the chamber. An average of five $R(\lambda)$ spectra, each with an integration time of 544 ms, was taken every minute during daytime (Malvern Panalytical, 2023). Every 10 minutes, the fiber optic was switched from one intertidal chamber to the other, in order to measure $R(\lambda)$ in both the treatment and control. Light conditions were controlled inside of the chambers and the reflectance calibration was performed each morning at the very first moment of low tide using a Spectralon white reference with 99% Lambertian reflectivity.

1.3.2 Spectrum post-processing

A Savitzky-Golay smoothing function with a 5 nm moving window was applied to each spectrum using the “hsdar” package in R (Lehnert et al., 2017). The second derivative at 665 nm ($\lambda = 665$), showing the highest variability between the control and the treatment, was tested as an indicator of the spectral changes following HWs.

The effect the HW on $R(\lambda)$ was also quantified using two radiometric indices: